

Motif by Motif: Controllable Batik Pattern Generation with Structure-Aware Diffusion

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ABSTRACT

Batik patterns follow rules: repeating motifs, isen fillers and borders that artisans combine with care. Generic image generators ignore these rules and produce patterns that look right from far away and wrong up close. We condition a diffusion model on a motif layout sketch and a filler vocabulary, trained on 2,300 photographed cloths documented with their makers. In a study with 11 artisans, generated drafts were rated useful as starting points in 64 percent of sessions.

Keywords: diffusion models, batik, cultural heritage, pattern generation, co-creation

1. Introduction

We did not want a machine that replaces artisans. We wanted a sketchbook that speaks their vocabulary, so a designer can say put a kawung field here and a parang border there and get a draft worth arguing about.

2. Method

Each training cloth is annotated with a coarse layout map: motif regions, filler regions and borders. The diffusion model receives the layout as extra channels and a filler vocabulary token per region. We tile generation with overlap so repeats line up at the seams [1, 2].

Every photographed cloth was documented with the maker's permission and credited in the dataset card.

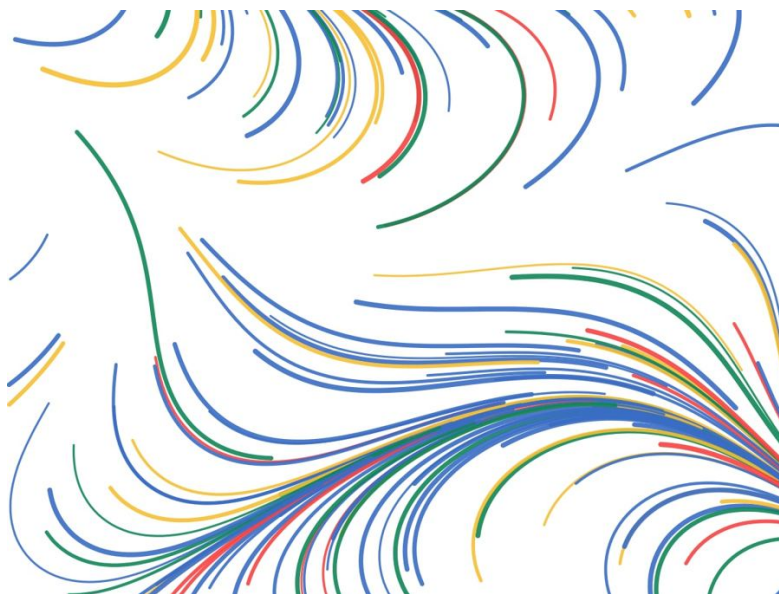


Figure 1. Flow of the layout field used to align repeating motifs across tile seams.

3. Results

Layout conditioning improves motif fidelity as judged by artisans and keeps repeats seamless (Table 1). Artisans mostly edited fillers and kept the generated layouts.

Model	Motif fidelity (1 to 5)	Seam errors	Useful drafts
Unconditioned	2.3	31%	18%
Text prompt only	2.9	22%	37%
Layout conditioned (ours)	4.1	4%	64%

Table 1. Artisan ratings and seam checks over 44 sessions.

4. Discussion

Several artisans asked for the opposite tool: take a finished cloth and suggest new fillers. That inversion is our next study. Questions about ownership of generated drafts came up in every session.

5. Conclusion

Structure-aware conditioning turns a generic generator into a useful co-design partner for batik.

References

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